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SUMMARY

Principal Software Engineer with 25+ years across .NET/backend, AI agent infrastructure & governance, cloud, and security. Recently building production AI agents and Model Context Protocol (MCP) servers, architecting ML pipelines for an AI startup, maintaining GayHydra (a security-hardened fork of NSA Ghidra), and architecting Scylla, a ground-up hexagonal reverse-engineering platform built on it. Deep C#/.NET and Python background and a clear technical writer. USMC veteran.

SELECTED OPEN-SOURCE & AI PROJECTS

GayHydra — hardened fork of NSA Ghidra C++, Java

- Sole architect of a security-hardened fork of NSA Ghidra (a multi-million-line C++/Java codebase) — drove a 42-item principal-architect audit into shipped releases across security, the decompiler, and CI.
- Hardened the Java-deserialization attack surface, modernized the decompiler's host–worker IPC into a versioned, schema-validated protocol, and shipped a multi-OS, Sigstore keyless-signed release pipeline with a CycloneDX SBOM.

Scylla — hexagonal reverse-engineering platform on the GayHydra engine Rust, Cap'n Proto, gRPC, MCP

- Architected and built a hexagonal reverse-engineering platform: GayHydra wrapped as a durable Rust core behind swappable adapters (an MCP server for AI agents first), with the model as a zero-copy Cap'n Proto artifact — the "narrow waist" that decouples the engine from every consumer.
- Designed a four-pass re-anchoring matcher that carries analyst annotations across recompiles and *cross-architecture* rebuilds (x86-64, aarch64, 32-bit) under a hard zero-wrong invariant, enforced by a ratcheted release gate over an 11-class binary corpus.
- Ran the engine as a sandboxed gRPC service — warm in-process runtime (about 3x faster), worker pool, no-egress container, keyless-signed artifacts — across 40+ logged architecture decisions, with the major bets de-risked by spikes first.

KaliMCP & PerplexityAgent Python

- Two security-hardened Model Context Protocol (MCP) servers exposing, respectively, a curated subset of Kali Linux security tools and the Perplexity Search/Sonar APIs to AI agents; per-invocation refuse-list guards and audit logging, hardened to NSA MCP security guidance.

omind Python

- Harness-agnostic memory & governance layer that gives every AI coding agent on a machine one shared, file-based long-term memory and a single policy-enforcement point — wired across Claude Code, Codex, and other harnesses so a rule learned under one enforces under all, with one-command reproducible setup and a local web app to browse and edit agent memory.
- Designed the "checks and balances" that keep agents honest: a per-turn gate that blocks tool use until the agent consults its memory, a relevance verifier that rejects off-topic consults (so the gate can't be cleared by reading an irrelevant note), deliberate opt-in for high-risk actions (e.g. publishing to a public remote), and a compliance audit trail of violations, repeat offenders, and verifier verdicts — all fail-open, so a broken hook never wedges the agent, with self-healing drift detection and a doctor that diagnoses and repairs the wiring.

Open-Source Contributions C#/.NET, Python, Java

- **Omnigent** (Python) — contributor to an open-source meta-harness for AI agents (a policy/sandbox layer over Claude Code, Codex, and custom agents); authored three merged P0 security fixes (PRs #158, #161, #163) hardening session input validation (RCE), OIDC login (identity spoofing), and policy fail-closed enforcement.

- **Libation** (C#/.NET) — contributor to the open-source Audible library manager; merged PR #1814 (locale-dependent unit-test fix) with additional feature PRs in review.
- **hermes-agent** (Python) — contributor to an open-source personal AI agent framework with persistent memory, scheduled autonomy, and multi-provider LLM routing with automatic fallback.
- **OWASP ZAP** (Java) — ongoing contributor to the OWASP flagship web application scanner.

EXPERIENCE

Ronin 48 Mar 2026 - Present
Principal Software Architect & ML Engineer (Contract) *Remote*

- Designing overall system architecture for an AI startup; defining technical standards, guiding technology selection, and ensuring scalable, production-ready software design across the platform.
- Implementing and operationalizing machine learning pipelines; guiding model selection and deployment strategy, and integrating ML capabilities into production systems.

Career Break Nov 2024 - Mar 2026
EMT-B Certification & Community Service *Minden, Nebraska*

- Completed Emergency Medical Technician (EMT-B) certification and joined Minden Volunteer Fire Department as probationary firefighter / EMT. Self-directed study in AI/ML systems.

Black Hills Information Security / Active Countermeasures Mar 2024 - Nov 2024
Senior Developer *Remote*

- Upgraded Windows Service from dotNET Framework 4.x to dotNet 8, refactored code to be testable and created unit tests. Built Windows ports of Unix software. Wrote documentation and tutorials for blog posts and webcasts.

CrowdStrike Aug 2022 - Mar 2024
Senior Technical Support Engineer *Remote*

- SME for SaaS and on-prem LogScale(Humio), Kafka, and log shippers. Created sandbox infrastructure using terraform and k8s. Wrote automation and tooling in bash and Python.

CloudHesive Aug 2021 - May 2022
Cloud Operations Supervisor / Cloud Support Engineer II *Remote*

- Promoted to lead the Cloud Operations team supporting 2000+ customer systems across AWS, Azure, and VMware — hired and developed engineers, and ran monitoring (Datadog, SumoLogic, NewRelic) and backup/recovery across CloudFormation, CloudWatch, EC2, EFS, ELB, IAM, RDS, S3, and VPC.

Reinke Manufacturing Company Dec 2015 - Feb 2021
Senior Developer *Remote*

- Led development of middleware connecting an AS/400 ERP to a WinForms sales application; built the ASP.NET MVC authentication/authorization system; stood up and maintained Azure infrastructure; evaluated and recommended third-party SAST tooling.

Earlier Experience 1998 - 2021

- **Tmutla** – Cybersecurity Consultant (2021), Grand Island, NE. Web-application penetration testing and client reporting.
- **Alpine Testing Solutions** – Software Developer (2014–2015), Remote. T-SQL data import/export and XML data-load tooling.
- **Orthman Manufacturing** – IT Specialist II (2012–2014), Lexington, NE. Led a physical-to-virtual Windows migration and built internal .NET applications.
- **Intellicom Computer Consulting** – Senior Engineering Technician (2009–2012), Kearney, NE. Installed and maintained Active Directory, File, and Exchange servers.

- **United States Marine Corps** – Information Specialist (2003–2008), Washington, DC. Operated and supported the USMC network; graduate of the USMC Information Assurance Manager Course.
- **Rackspace** – Linux Systems Administrator (2003), San Antonio, TX.
- **Cimarron Software Services** – Systems Analyst at NASA Johnson Space Center (2001–2002), Houston, TX.
- **MegaHaus** – Help Desk Technician (1998), Dickinson, TX.

TECHNICAL STRENGTHS

AI/ML	Model Context Protocol (MCP) servers, agentic tooling & orchestration, tool-use / function calling, Claude Code & Codex, LangChain, LangGraph, Hugging Face, OpenAI & Anthropic APIs, Ollama, RAG pipelines, vector databases (pgvector, Pinecone), prompt engineering
Cloud	AWS (EC2, S3, RDS, IAM, VPC, ELB, EFS, CloudFormation, CloudWatch), Azure, Terraform
Tools	Burp Suite, Postman, Wireshark, NMap, LogScale (Humio), Kafka, Docker, Kubernetes
Languages	Python, C#, TypeScript, JavaScript, VB.NET, T-SQL, PowerShell, Bash
Certifications	SSCP, AWS AI Practitioner, AWS Solutions Architect Associate, AWS Cloud Practitioner, CompTIA A+, CompTIA Network+, MS: Programming in C#
Memberships	OWASP, ACM, DC402, RITSEC, SPARSA.ORG, OperationCode

EDUCATION

Eastern University 2026 – Present
MS, Applied Artificial Intelligence (In Progress)

University of Maryland University College
Bachelor of Science, Computer and Information Science

Rochester Institute of Technology
MicroMaster's in Cybersecurity

Graduate & Professional Certificates
Computer Science (Kennesaw State); IT Project Management (Colorado State); Unix System Administration (UMUC)

VOLUNTEER EXPERIENCE

Community & Industry Service 2021 - Present

- **Minden Volunteer Fire Department** — Probationary Firefighter / EMT-B (Dec 2024 – Present), Minden, NE.
- **Women in CyberSecurity (WiCyS)** — Student Mentor (2021–2024).
- **Army National Guard** — Red Team (OPFOR) Lead, Cybershield 2021.